



BOARD OF INTERMEDIATE & SECONDARY EDUCATION, LARKANO

Established to serve affiliated schools and colleges across the Larkano region

Higher Secondary School Certificate
Examination Syllabus

CHEMISTRY

Grades XI – XII

This syllabus will be examined in both Annual and Re-sit Examination sessions beginning with the Annual Examinations 2023.

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Preface

The Board of Intermediate & Secondary Education, Larkano (BISE Larkano) is a public examination body responsible for the conduct of Secondary School Certificate (SSC) and Higher Secondary School Certificate (HSSC) examinations for its affiliated schools and colleges. The Board is committed to developing examination syllabi that promote conceptual understanding and scientific reasoning, and that remain aligned with the national curriculum and with contemporary international standards in science education.

This document sets out the syllabus for Chemistry at the HSSC level (Grades XI and XII). It has been developed in recognition of chemistry's role as a foundation for further study and employment in medicine, engineering, pharmacy, environmental science and a wide range of other scientific and industrial fields.

The syllabus was developed following consultation with subject specialists, teachers of affiliated institutions, and reviewers with expertise in chemistry education, and reflects a deliberate effort to move examination preparation away from rote memorisation of facts and toward the disciplined observation, calculation and reasoning that underpin the subject.

We are confident that this syllabus will support teachers and students of affiliated institutions in building a rigorous and practical understanding of chemistry, and we wish every success to those who study and teach it.

Controller of Examinations

Board of Intermediate & Secondary Education, Larkano

Understanding of BISE Larkano Syllabi

1. The BISE Larkano syllabi guide students, teachers, parents and other stakeholders regarding the topics that will be taught and examined in each grade (XI and XII). In each syllabus document, the content progresses from simple to complex, facilitating gradual, conceptual learning.
2. The topics of each syllabus are divided into sub-topics and Student Learning Outcomes (SLOs). The sub-topics and SLOs define the depth and breadth at which each topic will be taught, learnt and examined.
3. Each SLO begins with an achievable, assessable command word such as ‘describe’, ‘apply’, ‘analyse’ or ‘evaluate’. Examination questions are framed using the same command words, or their equivalents, to elicit evidence of these competencies in students’ responses.
4. The SLOs are classified under three cognitive levels: Knowledge (K), Understanding (U), and Application and other higher-order skills (A), to support effective planning by teachers and the balanced construction of examination papers.
5. By focusing on the achievement of SLOs rather than on the reproduction of memorised facts, this syllabus aims to encourage students to reason scientifically, to justify their conclusions with evidence and calculation, and to communicate their thinking clearly — skills that extend well beyond a single assignment or examination.
6. While the syllabus recommends locally available textbooks and resources for achieving these outcomes, teachers and students are encouraged to draw on multiple teaching and learning resources, and to acknowledge any resource that materially shapes a solution submitted for assessment.

Subject Rationale of Chemistry

What will you learn in Chemistry?

Chemistry is the study of matter, its composition, its properties and the changes it undergoes — knowledge that underlies medicine, agriculture, energy, materials and much of modern industry. This syllabus teaches students to explain observable phenomena in terms of atoms, bonds and energy, and to support their explanations with careful measurement and calculation rather than with memorised statements.

Many students who can recall a definition have not yet learned to apply it — to predict how a system at equilibrium will respond to a change in conditions, to work out the shape of a molecule from first principles, or to interpret experimental data as evidence for a reaction mechanism. This syllabus addresses that gap directly. Students will study the structure of the atom, the behaviour of gases, liquids and solids, the principles governing chemical equilibrium, kinetics, thermochemistry and electrochemistry, and the foundations of organic chemistry.

Throughout, the syllabus emphasises the application of chemical principles to unfamiliar situations, and the ability to reason from evidence — in the laboratory and in the examination room — rather than the recall of isolated facts.

Where will it take you?

The foundation provided by this syllabus prepares students for further study in Chemistry, Medicine, Pharmacy, Chemical Engineering and related disciplines, and for a wide range of careers, including:

- Doctor / Pharmacist
- Chemical Engineer
- Biochemist
- Environmental Scientist
- Materials Scientist
- Forensic Analyst
- Quality Control / Quality Assurance Chemist
- Petrochemical or Process Industry Specialist

How to approach the syllabus?

The concept map on the following page gives an overview of the entire syllabus. The topics and Student Learning Outcomes guide what has to be achieved, and the Scheme of Assessment explains how each topic will be weighted and examined.

Concept Map

The concept map below organises the syllabus into four connected strands, all resting on a foundation of Measurement, Stoichiometry and Scientific Reasoning — the quantitative and logical skills that make every other strand of chemistry possible.

Structure of Matter

- Atomic Structure and Quantum Numbers
- Periodic Classification of Elements
- Chemical Bonding and Molecular Shape
- The Mole Concept and Stoichiometry

States and Behaviour of Matter

- Gases and the Gas Laws
- Liquids and Intermolecular Forces
- The Solid State
- Chemical Equilibrium and Acid-Base Behaviour

Chemical Change and Energy

- Chemical Kinetics
- Thermochemistry
- Electrochemistry
- Reaction Mechanisms

Organic and Applied Chemistry

- Functional Groups and Nomenclature
- Hydrocarbons
- Chemistry in Industry and Everyday Life
- Laboratory Investigation and Data Analysis

Measurement, Stoichiometry and Scientific Reasoning underpin all four strands: every calculation, every prediction and every laboratory observation in this syllabus is ultimately a tool for reasoning carefully from evidence to a defensible conclusion.

Student Learning Outcomes of Chemistry

Part I (Grade XI)

1. Basic Concepts of Chemistry

Topics and Sub-topics	Student Learning Outcomes	K	U	A
1.1 Matter and Measurement	describe the major branches of chemistry and give examples of their application in everyday life;	✓		
	distinguish between a physical change and a chemical change in matter;		✓	
	express the result of an experimental measurement to an appropriate number of significant figures;			✓
1.2 The Mole Concept and Stoichiometry	define the mole and state the value of the Avogadro constant;	✓		
	calculate the molar mass, number of moles or number of particles present in a given sample;			✓
	determine the empirical and molecular formula of a compound from experimental composition data;			✓
	solve stoichiometric problems involving a limiting reagent and percentage yield;			✓

2. Atomic Structure

Topics and Sub-topics	Student Learning Outcomes	K	U	A
2.1 Development of Atomic Models	outline the historical development of atomic models, from Dalton's theory to the quantum mechanical model;	✓		
	describe the significance of atomic emission spectra to the development of atomic theory;		✓	
2.2 Quantum Numbers and Electronic Configuration	describe the four quantum numbers and the information each provides about an electron in an atom;	✓		
	write the electronic configuration of a given atom or ion using the Aufbau principle, the Pauli exclusion principle and Hund's rule;			✓
	relate the electronic configuration of an element to its position in the periodic table;		✓	

3. Gases

Topics and Sub-topics	Student Learning Outcomes	K	U	A
3.1 The Gas Laws	state Boyle's law, Charles's law and Avogadro's law;	✓		
	apply the general gas equation to solve problems involving the pressure, volume, temperature and amount of a gas;			✓
	explain the assumptions of the kinetic molecular theory of gases;		✓	
3.2 Real Gases	distinguish between the behaviour of an ideal gas and a real gas;		✓	
	explain the conditions of temperature and pressure under which a real gas deviates most from ideal behaviour;		✓	

4. Liquids and Solids

Topics and Sub-topics	Student Learning Outcomes	K	U	A
4.1 Intermolecular Forces and Liquids	describe the types of intermolecular forces present in a liquid;	✓		
	relate the physical properties of a liquid, e.g. boiling point, viscosity and surface tension, to the strength of its intermolecular forces;		✓	
4.2 The Solid State	distinguish between crystalline and amorphous solids;		✓	
	classify a solid according to the type of particle and bonding it contains, i.e. ionic, covalent, molecular or metallic;	✓		
	relate the physical properties of a solid to its underlying structure and bonding;			✓

5. Chemical Equilibrium

Topics and Sub-topics	Student Learning Outcomes	K	U	A
5.1 The Equilibrium State	describe the characteristics of a chemical system at dynamic equilibrium;	✓		
	write the equilibrium constant expression for a given reversible reaction;			✓
	apply Le Chatelier's principle to predict the effect of a change in concentration, pressure or temperature on a system at equilibrium;			✓
5.2 Acid-Base Equilibria	describe the Bronsted-Lowry theory of acids and bases;	✓		
	calculate the pH of a solution from its hydrogen ion concentration;			✓

Part II (Grade XII)

6. Chemical Kinetics

Topics and Sub-topics	Student Learning Outcomes	K	U	A
6.1 Rate of Reaction	define the rate of a chemical reaction;	✓		
	describe the factors that affect the rate of a chemical reaction;		✓	
	interpret a rate law and determine the order of a reaction from experimental data;			✓
6.2 Reaction Mechanisms	explain the concept of activation energy and the role of a catalyst with reference to a reaction-energy profile;		✓	
	relate a proposed reaction mechanism to an experimentally determined rate law;			✓

7. Thermochemistry

Topics and Sub-topics	Student Learning Outcomes	K	U	A
7.1 Energy Changes in Reactions	distinguish between an exothermic and an endothermic reaction;		✓	
	define enthalpy change and the standard enthalpy of a reaction;	✓		
	calculate the enthalpy change of a reaction by applying Hess's law;			✓
7.2 Applications of Thermochemistry	calculate the enthalpy change of a reaction from given bond energies;			✓

8. Electrochemistry

Topics and Sub-topics	Student Learning Outcomes	K	U	A
8.1 Oxidation and Reduction	assign oxidation states to the atoms in a given compound or ion;			✓
	balance a redox equation using the ion-electron (half-reaction) method;			✓
8.2 Electrochemical Cells	describe the construction and operation of a galvanic (voltaic) cell;	✓		
	calculate the standard cell potential of a galvanic cell from standard electrode potentials;			✓
	distinguish between a galvanic cell and an electrolytic cell;		✓	

9. Periodic Classification and Chemical Bonding

Topics and Sub-topics	Student Learning Outcomes	K	U	A
9.1 Periodicity of Properties	describe the general trends in atomic radius, ionisation energy and electronegativity across a period and down a group;		✓	
	explain periodic trends in terms of nuclear charge and electron shielding;			✓
9.2 Chemical Bonding	describe ionic, covalent and coordinate covalent bonding with reference to the transfer or sharing of electrons;	✓		
	use the valence-shell electron-pair repulsion (VSEPR) theory to predict the shape of a simple molecule or ion;			✓
	relate the polarity of a molecule to its shape and to the electronegativity of its constituent atoms;			✓

10. Introduction to Organic Chemistry and Hydrocarbons

Topics and Sub-topics	Student Learning Outcomes	K	U	A
10.1 Fundamentals of Organic Chemistry	describe the tetravalency of carbon and its significance for the diversity of organic compounds;	✓		
	classify an organic compound according to its functional group;		✓	
	name a simple organic compound using IUPAC nomenclature;			✓
10.2 Hydrocarbons	distinguish between saturated and unsaturated hydrocarbons;		✓	
	describe a characteristic chemical reaction of an alkane, an alkene and an alkyne;			✓
	relate the properties of hydrocarbons to their use as fuels and as industrial feedstocks;			✓

Scheme of Assessment

Each grade of Chemistry is examined through a Theory paper and a Practical (laboratory) examination, whose marks combine to the 100 marks recorded for the subject on the BISE Larkano Marks Certificate. The Theory paper carries 85 marks and the Practical examination carries 15 marks, matching the maximum marks shown against ‘Chemistry (TH)’ and ‘Chemistry (PR)’ for each year on the certificate.

Grade XI

Table 1: Number of Student Learning Outcomes by Cognitive Level

Topic No.	Topic	No. of Sub-topics	K	U	A	Total
1	Basic Concepts of Chemistry	2	2	1	4	7
2	Atomic Structure	2	2	2	1	5
3	Gases	2	1	3	1	5
4	Liquids and Solids	2	2	2	1	5
5	Chemical Equilibrium	2	2	0	3	5
	Total	10	9	8	10	27
	Percentage		33%	30%	37%	100%

Table 2: Theory Exam Specification — Grade XI — Chemistry (TH)-I — 85 Marks

Topic No.	Topic	MCQs (Marks)	CRQs (Marks)	Total Marks
1	Basic Concepts of Chemistry	5	10 (1 CRQ)	15
2	Atomic Structure	6	11 (1 CRQ)	17
3	Gases	5	10 (1 CRQ)	15
4	Liquids and Solids	6	11 (1 CRQ)	17
5	Chemical Equilibrium	8	13 (1 CRQ)	21
	Total	30	55	85

Table 2A: Overall Marks Distribution — Grade XI

Component	Certificate Entry	Marks
Theory Examination	Chemistry (TH)-I	85
Practical Examination	Chemistry (PR)-I	15
Total		100

Grade XII

Table 3: Number of Student Learning Outcomes by Cognitive Level

Topic No.	Topic	No. of Sub-topics	K	U	A	Total
6	Chemical Kinetics	2	1	2	2	5
7	Thermochemistry	2	1	1	2	4
8	Electrochemistry	2	1	1	3	5
9	Periodic Classification and Chemical Bonding	2	1	1	3	5
10	Introduction to Organic Chemistry and Hydrocarbons	2	1	2	3	6
	Total	10	5	7	13	25
	Percentage		20%	28%	52%	100%

Table 4: Theory Exam Specification — Grade XII — Chemistry (TH)-II — 85 Marks

Topic No.	Topic	MCQs (Marks)	CRQs (Marks)	Total Marks
6	Chemical Kinetics	5	10 (1 CRQ)	15
7	Thermochemistry	5	9 (1 CRQ)	14
8	Electrochemistry	6	11 (1 CRQ)	17
9	Periodic Classification and Chemical Bonding	7	13 (1 CRQ)	20
10	Introduction to Organic Chemistry and Hydrocarbons	7	12 (1 CRQ)	19
	Total	30	55	85

Table 4A: Overall Marks Distribution — Grade XII

Component	Certificate Entry	Marks
Theory Examination	Chemistry (TH)-II	85
Practical Examination	Chemistry (PR)-II	15
Total		100

- A Multiple Choice Question (MCQ) requires candidates to choose one best/correct answer from four options. Each MCQ carries ONE mark.
- A Constructed Response Question (CRQ) requires students to respond with a structured explanation, a worked calculation, a labelled diagram, or a chemical equation, as appropriate to the SLO being assessed.
- There will be two Theory examinations, one at the end of Grade XI (Chemistry (TH)-I) and one at the end of Grade XII (Chemistry (TH)-II). Each Theory paper carries 85 marks, is of 3 hours' duration, and consists of Paper I (MCQs) and Paper II (CRQs).
- There will be two Practical examinations, one at the end of Grade XI (Chemistry (PR)-I) and one at the end of Grade XII (Chemistry (PR)-II). Each Practical examination carries 15 marks and requires students to carry out a volumetric, qualitative or physical-chemistry experiment under supervised laboratory conditions, and to record and interpret their own results.

- The Theory and Practical marks for a given year are reported separately on the BISE Larkano Marks Certificate and are combined only in the year's subject total ($85 + 15 = 100$). A candidate must attempt both components; a Theory or Practical mark cannot substitute for the other.
- Tables 1 and 3 indicate the number and nature of SLOs in each topic and are provided as a guide to the construction of examination papers; they do not translate directly into marks. Emphasis has deliberately been placed on Application and other higher-order skills, to discourage rote memorisation of facts.
- Where a CRQ requires a numerical answer, full credit requires a correctly set-out method and appropriate units, not merely the final value.

Academic Integrity and Laboratory Safety

All work submitted for assessment in Chemistry — whether a written answer, a calculation, or a laboratory report — must represent the individual, honest effort of the student submitting it. Affiliated institutions are expected to uphold the BISE Larkano Academic Integrity Policy in full, and to make students aware of the following subject-specific guidance.

Collaboration and Use of Sources

Students are encouraged to discuss chemical concepts, experimental observations and problem-solving strategies with their peers, and to consult the prescribed textbook and other reputable resources to build their understanding. However, copying a solution, a calculation, or a laboratory report from any source — including another student, a book, or an online resource — and submitting it as one's own, with or without superficial changes, is a serious violation of academic integrity.

Integrity of Experimental Data

Where practical work forms part of a student's assessment, all recorded observations and readings must be genuine. Fabricating, adjusting or selectively omitting experimental data to obtain an expected or more convenient result is a serious violation of academic integrity, regardless of how small the discrepancy may seem.

Laboratory Safety

Affiliated institutions are responsible for ensuring that all practical work is conducted under appropriate supervision, with attention to the safe handling, storage and disposal of chemicals, and with students made aware of relevant hazard information before an experiment begins. A student's disregard for reasonable safety instructions during practical work may be treated as a conduct matter independent of, and in addition to, their academic assessment.

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